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Module 4: Software Design
Session 1: Design Approaches and Requirements Analysis

Software Developer Level 4



What you will be able to do

By the end of this session you will be able to:

1. Analyse a client brief and request data to identify requirements, constraints and open questions.
2. Compare four software design approaches and justify which one best fits each part of a project.
3. Produce a domain concept map and a task flow that communicate the problem to stakeholders.

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The module: one design, built over six sessions

Three days, two sessions a day. Each session produces something you keep.



Approaches and requirements
Concept map, task flows, open questions

UI and UX
Wireframes and a clickable prototype

Use cases and stories
Use case, stories and test scenarios

Architecture and UML
Class, sequence and swimlane diagrams

Secure design, patterns
A reviewed design and a decision record

Design review
Your own design pack, shown to a review board

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Our client for the module: Kestrel Moor Homes

A housing association with about 6,800 homes and about 21,000 repair requests a year

Tenants report repairs by phone, web form or email, or through a housing officer.

The board has approved a Tenant Repairs Portal. Design runs October to December 2026; build follows in 2027.

You are the design team for the whole module.

Fictional organisation and synthetic data.



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Your Reflective Log and your Session 6 project

Reflective Log

Reflective Log.docx is in the Module files folder.

At the end of each session, write one short entry about something you made. It takes about 5 minutes.

Copy the key points into your OneFile timesheet reflection.

Bring a project to Session 6

In Session 6 you extend the design of your own project and present it to a review board.

Bring your Software Foundations project plan, or a workplace feature described without real data.

No project, or one too sensitive to show? Use the fallback brief in Case Brief Part 6.

No personal or customer data, credentials or confidential material, and no employer data in AI tools your employer has not approved.

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Design approaches

Where design sits in the life cycle, four ways to approach it, and how to choose between them with evidence.

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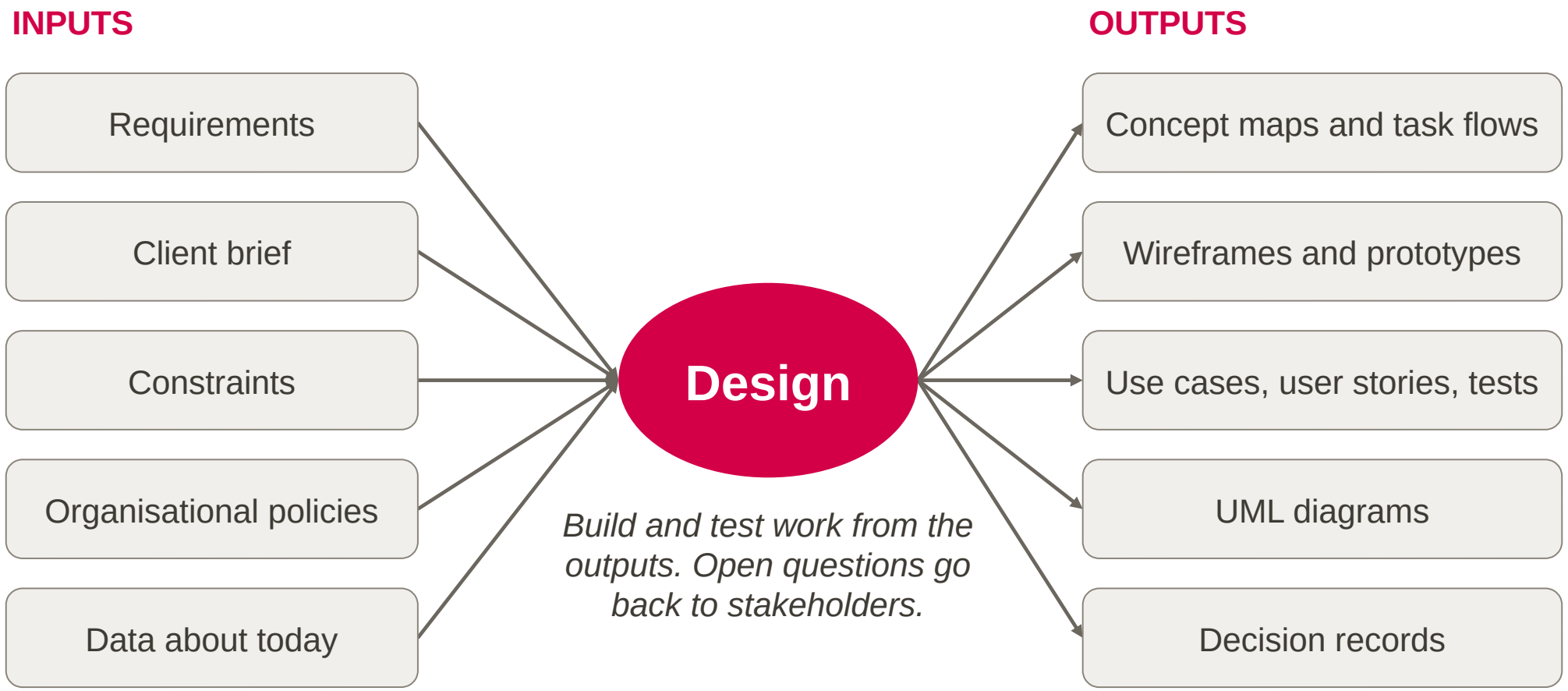
Before the first line of code

A developer starts building the repairs portal tomorrow. What do they need from design first?

Your Prepare question. Type one thing in the chat: we will sort the answers into what design takes in and what it hands on.

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What design takes in and hands on



Four design approaches

Each approach decides what you design first and how you find out if it is right.



User-centred

Start from the people who will use it: research, design, test with them.

Iterative

Build a thin slice, get feedback, adjust, and repeat.

Component-based

Split the system into parts with clear interfaces.

Domain-driven

Model the business rules with the experts, in one shared language.

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Four approaches compared

Approach	Typical first artefacts	Fits when	Watch out for
User-centred	Personas, task flows, prototypes tested with users	Users vary and their needs are unclear	Research takes time; users cannot judge the back end
Iterative	Release plan for a thin slice, with a feedback measure	Priorities are uncertain and feedback is quick to get	Slices drift without a direction; rework adds up
Component-based	Component diagram with named interfaces	Parts connect to systems or suppliers that may change	Interfaces need agreeing early; many parts add overhead
Domain-driven	Concept map and glossary agreed with experts	Rules are complex and terms clash between teams	Needs experts' time; heavy for a simple system

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Design approach or programming paradigm?

You met paradigms in Applied Programming. This module is about design approaches.

PROGRAMMING PARADIGM

How you structure the code

Procedural, object-oriented, event-driven, functional

Chosen per language or module

You see it in classes, functions and handlers

DESIGN APPROACH

How you decide what to design, in what order, and how you check it

User-centred, iterative, component-based, domain-driven

Chosen per part of the system, from evidence

You see it in personas, release plans, interfaces and glossaries

Choosing with evidence: five questions

Ask these of each part of the system, and back every answer with evidence.

1. **How clear are the requirements?**
   Unclear or disputed needs point to user research and early testing.
2. **How complex is the domain?**
   Many rules, special cases and clashing terms point to modelling the domain.
3. **What is likely to change?**
   A supplier or system that may be replaced points to parts behind interfaces.
4. **How involved can stakeholders be?**
   Frequent access to users and experts makes short feedback cycles possible.
5. **What must be documented?**
   Regulated or audited work needs decisions written down, whatever the approach.

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Worked example: a leisure-centre booking app

BRIEF EXTRACT: RIVERSIDE LEISURE CENTRE

(1) Members aged 16 to 90 book fitness classes; many say the current app is confusing.

(2) The payment provider's contract ends next year and a new provider will be chosen.

(3) The manager wants online booking for one class type by January, then to add more.

(4) Booking rules are simple: one place per member per class; cancel up to 2 hours before.

For each part: the approach, the evidence for it and the first artefact.

1. Booking journey: user-centred. Wide age range, a 'confusing' app. First: a task flow tested with members.
2. Payments: component-based. The provider will change. First: the payment interface.
3. Delivery: iterative. One class type, then more. First: a release plan for slice one.
4. Rules: domain-driven not needed. Few, stable rules; a short glossary is enough.

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Activity 1: Choose the approach

Pairs (2 minutes on your own first)

Choose and justify an approach for four parts of the portal.

1. On your own for 2 minutes: read situations A to D (Supporting Material section 4).
2. As a pair, pick an approach for each: user-centred, iterative, component-based, domain-driven.
3. Quote the evidence that decided it: a line from the brief or a figure from the data.
4. Name one approach you rejected for each situation, and why the evidence rules it out.

Extension: see the Activities document.

TIME
20 minutes working time

GROUPING
Pairs (2 minutes on your own first)

YOU NEED
Situations A to D (Supporting Material section 4)
Activity 1 table (Activities document)

OUTPUT AND SUCCESS
Four rows: approach, the evidence that decided it, an approach rejected and why.

Debrief: Which situation could you defend two ways, and what settled it?

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Break

20 minutes

When we come back: reading a brief like a designer, then the main task of the session.

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Reading a brief like a designer

What to pull from a brief and the data, and how to show it as a concept map and a task flow.

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Seven things to pull from a brief

BRIEF EXTRACT: RIVERSIDE LEISURE CENTRE

(1) Members book classes in the app or at reception.

(2) They want to book in seconds and know they have a place.

(3) Each class has a studio, an instructor and up to 20 places.

(4) A member may hold three bookings at once.

(5) The new app must use the existing membership database.

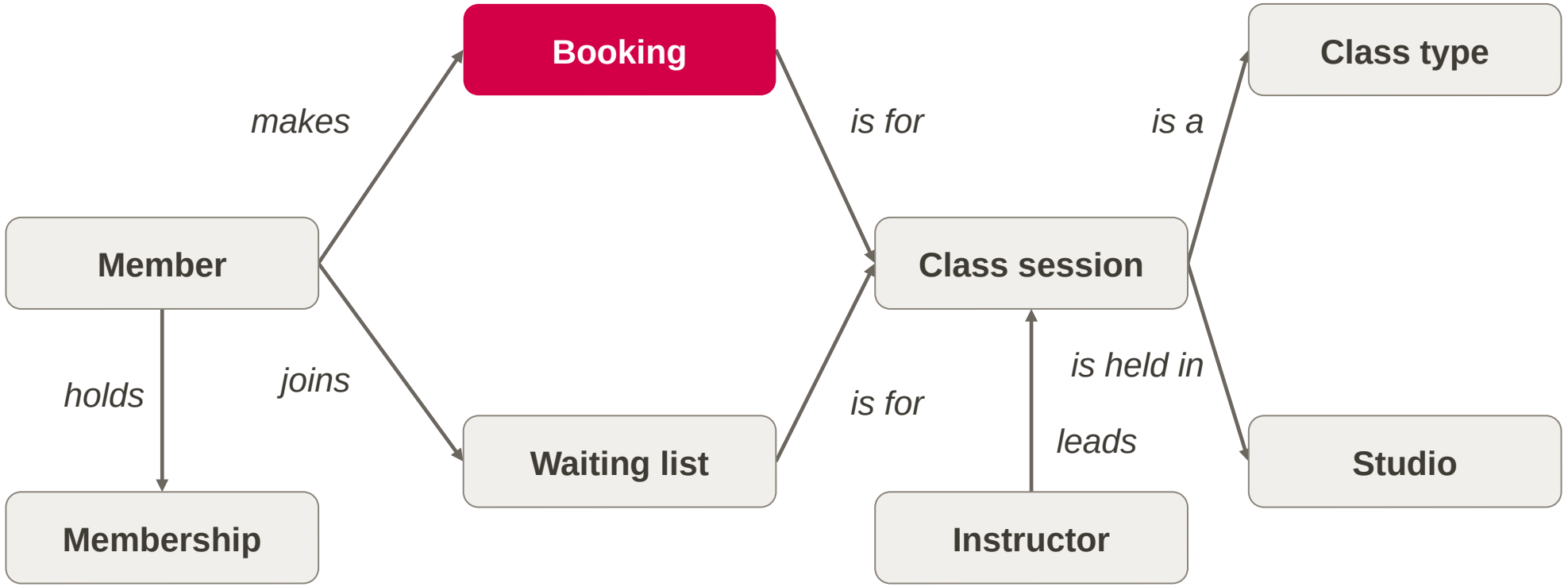
(6) Full classes leave members ringing reception about gaps.

(7) Waiting-list places are 'offered', but for how long?

1. Actors: who does something
2. Goals: what each actor wants
3. Things: nouns the system must know
4. Rules: what must always or never happen
5. Constraints: limits on how you build
6. Pain points: where it goes wrong today
7. Open questions: what to ask, and whom

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Worked example: a leisure-centre concept map



Nouns in boxes and a verb on every line, so each link reads as a sentence.

Worked example: a task flow for booking a class

The member's steps, in order, from the member's point of view.



Member is signed in

Sees times and spaces left

No: join the waiting list instead

One tap

Reminder 2 hours before

The flow ends when the member's goal is met. Today the same member rings reception and waits on hold.

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Live demo: draw.io in five moves

In draw.io

1. Open app.diagrams.net, then File, Open from, Device, and choose the starter file.

2. Drag a rounded rectangle from the General shapes and type a concept.

3. Hover over a shape, drag its arrow to another shape, then double-click the line to add a verb.

4. Switch pages with the tabs at the bottom: Concept map, Task flow, Open questions.

5. File, Save as, choose Device. Keep the .drawio ending.

No draw.io?

Use PowerPoint shapes and connectors, one slide per page.

Or draw on paper, photograph it and post it in the class channel.

Your tutor looks at the content, so a clear photograph is fine.

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The request data: what actually happens



150 requests, 1 September to 12 October 2026

Turn on the header filters. Start with four columns:

category: the repair type chosen when it was logged

priority_assigned: Emergency, Urgent or Routine, set at triage

status: Completed or Open on 16 October 2026

tenant_description: the problem in the tenant's own words

No names: tenant and property references are pseudonymous.

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Worked example: check a claim against the data

20.2

hours: mean wait from report to triage

Claim (Part 1): housing officers log repairs "usually the next working day".

Filter channel to Housing officer: 14 requests.

Mean wait 20.2 hours; longest 33.6 hours (sort the column).

Verdict: the data is consistent with the brief. Note the filter and the column.

Open questions: who to ask, and why

METHOD

> **A useful open question names who can answer it and quotes the evidence that made you ask.**

Leisure centre: "To the bookings manager: how long does a member have to accept a waiting-list place? Evidence: the brief says places are 'offered' but gives no time limit."

Opinions ("Is the app good?") and assumptions ("Members will not mind waiting") are not open questions.

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Activity 2: Read the brief, map the domain

Pairs to discuss, each learner draws

Map the repairs domain from the brief and the data.

1. On your own, scan Case Brief Parts 1 and 2. Fill the step 1 table, then compare with your partner.
2. Filter category, priority_assigned and status; read tenant_description. Fill the step 2 table.
3. Draw about 10 concepts on the Concept map page, with a verb on every line.
4. On the Task flow page, draw how a tenant reports a repair through the portal.
5. On the Open questions page, write two evidenced questions. Fill the step 5 table.

Extension: see the Activities document.

TIME
45 minutes working time

GROUPING
Pairs to discuss, each learner draws

YOU NEED
Case Brief, Parts 1 and 2
Repair Requests workbook or CSV
Session 1 starter file in draw.io

OUTPUT AND SUCCESS
Concept map, portal task flow, two evidenced open questions, three short tables.

Debrief: Which open question would change the design most if the answer surprised you?

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Live check

CHECK

1. Which of these is an output of design for a parents' evening booking app?

A The rule that each appointment slot lasts five minutes
B Last year's booking records showing which slots filled first
C A task flow showing how a parent books three teachers in a row
D The school's brief asking for online booking by the spring term

2. Permit app: uptake unknown; one ward can go first. Which choice has evidence?

A Domain-driven, because parking permits have their own vocabulary
B Iterative, because one ward can test uptake before it widens
C Component-based, because every modern app should use components
D Object-oriented, because residents and permits map neatly to classes

3. Which dentist-booking note is an open question rather than an assumption?

A Patients will stop phoning once online booking is live.
B Emergency slots can be booked online in the same way as check-ups.
C Reception staff already know which patients need longer slots.
D Practice manager: how many emergency slots are held back each day?

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This afternoon: from task flow to screens

Which step in your portal task flow will be hardest to make simple on a phone screen, and why?

Keep your draw.io file. In Session 2 your portal task flow becomes wireframes and a clickable prototype.

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Skills covered today

KSB and official wording	In plain English	Where we covered it today
Part of K1: the inputs and outputs (of the design stage)	What design takes in from requirements and hands on to the build and test stages	Where design sits; Activity 2; live check
Part of K7: software design approaches	Choosing between user-centred, iterative, component-based and domain-driven design	Activity 1; live check
B2: Applies logical thinking	Giving clear reasons, from evidence, for each design choice	Activities 1 and 2

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Skills covered today (continued)

KSB and official wording	In plain English	Where we covered it today
Part of B8: Shows curiosity to the business context in which the solution will be used; displaying an inquisitive approach to solving the problem	Asking what Kestrel Moor and its tenants really need before designing	Activity 2

What you did today

1. Design turns needs, constraints, policies and data into artefacts others build from.
2. Choose an approach per part of the system, not one for everything.
3. Every choice needs evidence: a quote, a rule, a requirement or a figure.
4. Concept maps show the problem in the client's words; task flows show the user's steps.
5. An open question names a stakeholder and quotes its evidence.

Before next time

Complete the Session 1 Consolidate Quiz (about 10 minutes).

Read the Session 2 Prepare document: UI and UX Design, Wireframes and Prototypes - Prepare.docx.

Keep your draw.io file safe: your portal task flow is where Session 2 starts.

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Reflective Log: your Session 1 entry

Complete the Session 1 entry in Reflective Log.docx, then copy key points into your OneFile timesheet reflection.

Today's prompt: the approach you chose for one part of the portal and why, and one open question for a stakeholder.

Add one line on how this could help in your own work.

Keep workplace examples general: no personal or customer data, credentials or confidential material, and no employer data in AI tools your employer has not approved.

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